Molecular basis of inheritance

The DNA

1. Read the following statements and choose the correct option.
Statement 1: A polymer of nucleotide formed has at one end a free phosphate moiety at 3'-end of sugar, which is referred to as 3'-end of polynucleotide chain.
Statement 2: At the other end of the polymer, the sugar has a free OH of 5'C group which is referred to as 5'-end of the polynucleotide chain.
1) Both statements are correct
2) Only statement 1 is correct but statement 2 is incorrect
3) Only statement 2 is correct but statement 1 is incorrect
4) Both statements are incorrect

2. Central dogma of molecular biology was proposed by
1) Francis Crick
2) James Watson
3) Ervin Chargaff
4) Watson and Crick

3. In a DNA molecule, two polynucleotide chains are different in
1) Number of purine and pyrimidine bases
2) Polarity
3) Molecular weight
4) All of the above

4. If there are 48×10^8 base pairs in a eukaryotic cell, then what will be the number of beads-on-string structure present in the nucleus?
1) 48×10^8
2) 24×10^6
3) 24×10^8
4) 2.4×10^8

5. Both deoxyribose and ribose belong to a class of sugars called
1) Trioses
2) Hexoses
3) Pentoses
4) Polysaccharides

6. What is incorrect for DNA?
1) The pitch of the helix is 3.4 nm.
2) Phosphate moiety is at 5' end of deoxyribose sugar.
3) Diameter of DNA is 2 nm.
4) The charge on the DNA is positive.

7. The force that holds DNA together in a double helix is
1) Hydrogen bonds
2) The force of the twist
3) N-glycosidic linkage
4) Ionic interactions

8. The fact that a purine base always pairs through hydrogen bonds with a pyrimidine base in the DNA double helix, leads to

1) The antiparallel nature
2) The semi-conservative nature
3) Uniform width throughout DNA
4) Uniform length in all DNA

9. The net electric charge on DNA and histones is
1) Both positive
2) Both negative
3) Negative and positive, respectively
4) Zero

10. The beads-on-strings is/are
1) Exons interrupted by introns
2) Nucleosomes in chromatin
3) Smaller submit of ribosomes in polysome
4) Amino acid in a long polypeptide chain

11. DNA double helix is
1) Complementary and parallel
2) Without supercoils
3) Complementary and antiparallel
4) Always linear

Search for Genetic Material and RNA World

12. A molecule that can act as a genetic material must fulfill the following criteria given below except
1) It should be able to generate its replica (Replication)
2) It should be chemically and structurally stable
3) It should provide the scope for slow changes (mutation) that are required for evolution
4) It should not be able to express itself in the form of 'Mendelian characters'

13. Griffith could distinguish the two strains of Streptococcus pneumoniae due to
1) Colony appearance in culture
2) Difference in their lethality in mice
3) Their size
4) Both (1) and (2)

14. Which of the following feature is represented by chemically less reactive and structurally more stable nucleic acid?
1) Catalytic nature
2) Direct involvement in protein synthesis
3) Scope for slow change
4) 2'-OH group

15. Identify the life process that evolved around RNA
1) Metabolism
2) Splicing
3) Translation
4) All of the above

16. Consider the following statements regarding transformation experiment.
(A) It was performed by F. Griffith on Streptococcus pneumoniae.
(B) It proved that DNA is the genetic material.
(C) Avery, Macleod, and McCarty worked to determine the biochemical nature of transforming principle.

(D) DNase inhibits the transformation.

The correct statement is/are

1) (A) and (B) only
2) (A) only
3) (A), (B), and (C) only
4) (A), (B), (C), and (D)

17. Read the following statements and choose the correct option.

Statement 1: The genetic material should be stable enough not to change with different stages of life cycle, age or with change in physiology of the organism.

Statement 2: Stability as one of the properties of genetic material was very evident in Griffith's "transforming principle' itself that heat, which killed the bacteria, at least did not destroy some of the properties of genetic material.

1) Only statement 2 is correct but statement 1 is incorrect
2) Only statement I is correct but statement 2 is incorrect
3) Both statements are correct
4) Both statements are incorrect

18. Read the following statements about Hershey and Chase experiment and choose the correct option.

I. Viruses grown. in the presence of radioactive phosphorus contained radioactive protein.
II. Viruses grown on radioactive sulphur contained radioactive DNA.
III. Bacteria which was infected with viruses that had radioactive DNA were radioactive, indicating that DNA was the material that passed from the virus to the bacteria
IV. Bacteria that were infected with viruses that had radioactive proteins were not radioactive. This indicates that proteins did not enter the bacteria from the viruses.

Correct statements are

1) All of the above
2) Only III and IV
3) Only I and II
4) Only I and III

19. Match the columns.

Column I		Column II	
A	Transformation	I	Miescher
B	Reverse central dogma	II	Griffith
C	Nuclein	III	Wilkins
D	X-ray diffraction data	IV	Temin and Baltimore

1) A-II; B–IV; C–I; D-III
2) A-IV; B–II; C–III; D-I
3) A-II; B-IV; C-III; D-I
4) A-I; B-III; C-IV; D-II

20. Teminism is concerned with _____ activity

1) Reverse transcriptase
2) DNA-dependent RNA polymerase
3) RNA-dependent DNA polymerase
4) Both (1) and (3)

21. Select the incorrect statement.

1) RNA was the first genetic material.
2) Some essential life processes like metabolism, translation, and splicing evolved around DNA.
3) DNA has evolved from RNA with chemical modification.

4) RNA acts as genetic material as well as catalyst.

22. Read the following statements and choose the correct option.

Statement I: In Hershey and Chase experiment, the bacteriophage attaches to the bacteria and its genetic material enters the bacterial cell.

Statement II: In Hershey and Chase experiment, the bacterial cell treats the viral genetic material as if it was its own and subsequently manufactures more virus particles.

Statement III: Meselson and Stahl worked to discover whether it was protein or DNA from the viruses that entered the bacteria.

Statement IV: Meselson and Stahl grew some viruses on a medium that contained radioactive phosphorus and some others on medium that contained radioactive sulphur.

1) All statements are correct
2) All statements are correct except I and II
3) All statements are correct except III
4) All statements are correct except III and IV

23. Read the following statements and choose the correct option

Statement 1: RNA being a catalyst was reactive and hence unstable.

Statement 2: DNA has evolved from RNA with chemical modifications that make it more stable.

1) Only statement 2 is correct but statement 1 is incorrect
2) Only statement 1 is correct but statement 2 is incorrect
3) Both statements are correct
4) Both statements are incorrect

24. Match the columns and select the correct option.

Column I (Scientist)		**Column II (Experimental material)**	
a	F. Griffith	1	Vicia faba
b	Meselson and Stahl	2	Bacteriophages
c	Taylor	3	Streptococcus pneumoniae
d	Hershey and Chase	4	Escherichia coli

	a	**b**	**c**	**d**
1)	1	2	3	4
2)	4	3	2	1
3)	3	4	1	2
4)	2	1	4	3

25. Which of the chemical groups makes RNA labile and easily degradable?

1) 3'-OH
2) 5'-PO_4
3) 2'-OH
4) 3'-PO_4

26. Match the columns and select the correct option.

Column I		**Column II**	
a	Double-helical model of DNA	1	Francis Crick
b	Transforming principle	2	Jacob and Monod
c	Central dogma	3	Frederick Griffith
d	Lac operon model	4	Waston and Crick

	a	b	c	d
1)	1	3	4	2
2)	4	3	1	2
3)	2	4	3	1
4)	3	1	2	4

27. RNA.is not a better genetic material because
1) It is chemically and structurally unstable
2) It is able to generate its replica
3) 2'-OH group present at every nucleotide in RNA as an additional moiety, which makes RNA labile and easily degradable.
4) Both (1) and (3)

DNA Replication and Transcription

28. Semiconservative mode of DNA replication was experimentally proved by Meselson and Stahl using
1) Heavy isotope of nitrogen, i.e., N^{15}
2) Experimental material with Vicia faba
3) Autoradiography technique
4) Radioactive sulphur and radioactive phosphorus

29. During DNA replication, the enzyme that unwinds the DNA helix is
1) Kornberg enzyme
2) Helicase
3) Topoisomerase
4) DNA polymerase III

30. DNA replication is
1) Semiconservative
2) Semi-discontinuous
3) Multidirectional
4) Both (1) and (2)

31. Discontinuous synthesis of DNA occurs in one strand, because
1) DNA molecule being synthesised is very long
2) DNA dependent DNA polymerase catalyses polymerisation only in one direction (5'→3')
3) It is a more efficient process
4) DNA ligase has to have a role

32. Read the following statements and choose the correct option.
Statement 1: The DNA polymerases can initiate the process of replication on their own.
Statement 2: The replication does not initiate randomly at any place in DNA.
1) Only statement 2 is correct but statement 1 is incorrect
2) Only statement 1 is correct but statement 2 is incorrect
3) Both statements are correct
4) Both statements are incorrect

33. DNA replication is
(A) Polymerization of nucleotides by DNA-dependent DNA polymerase activity
(B) Energetically very expensive process
(C) Polymerization of ribonucleotides in 5' → 3' direction
1) Only (A) is correct.
2) Only (A) and (B) are correct.
3) Only (A) and (C) are correct.
4) (A), (B), and (C) are correct.

34. A failure in cell division after DNA replication results into
1) Gene mutation
2) Chromosomal aberration

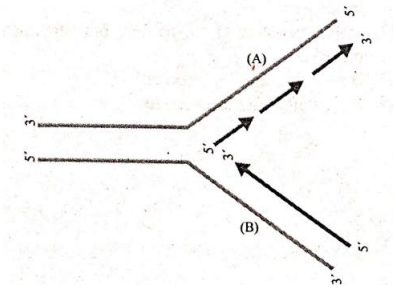
3) Polyploidy
4) All except (3)

35. Read the following statements about Meselson and Stahl and choose the correct option.
Statement 1: The DNA that was extracted from the culture two generations after the transfer from ^{15}N to ^{14}N medium had a hybrid or intermediate density.
Statement 2: DNA extracted from the culture after three generations was composed of equal amounts of this hybrid DNA and of 'light' DNA.
1) Only statement 2 is correct but statement 1 is incorrect
2) Only statement 1 is correct but statement 2 is incorrect
3) Both statements are correct
4) Both statements are incorrect

36. Semiconservative mode of DNA replication was experimentally proved in E.coli by
1) A. Kornberg
2) Benzer
3) Taylor
4) Meselson and Stahl

37. The following is generally used for creating density gradient during centrifugation.
1) NaCl
2) KCI
3) CsC1
4) $MgCl_2$

38. Study the given diagram, and find out the incorrect statement.



1) Unwinding of DNA molecule into two strands results in the formation of Y-shaped structure called replication fork.
2) At (A) template strand, the replication is discontinuous.
3) The nucleotide at 3' end is always the most recently added nucleotide to the chain.
4) At (B) template strand, newly formed DNA strand is called lagging DNA strand.

39. Read the following statements and choose the correct option.
Statement 1: DNA-dependent DNA polymerase enzymes are highly efficient enzymes as they have to catalyse polymerisation of a large number of nucleotides in a very short time.
Statement 2: Not only do these polymerases have to be fast, but they also have to catalyse the reaction with high degree of accuracy.
1) Only statement 2 is correct but statement 1 is incorrect
2) Only statement 1 is correct but statement 2 is incorrect
3) Both statements are correct
4) Both statements are incorrect

40. Isotopes used to prove semiconservative mode of DNA replication of DNA are
1) N^{14}, C^{14}
2) S^{35}, P^{32}
3) N^{14}, N^{15}
4) N^{14}, P^{32}

41. Read the given statements with respect to DNA replication.

A. DNA-dependent DNA polymerase catalyzes the polymerization in only one direction that is 5'→3'.

B. In eukaryotes, the replication of DNA takes place at the M phase of the cell cycle.

C. The discontinuously synthesized fragments are later joined by the enzyme ligase.

D. Deoxyribonucleoside triphosphate serves dual purpose. In addition to acting as substrates, they provide energy for polymerization reaction during DNA replication

How many of the above statements are correct?

1) Two
2) One
3) Four
4) Three

42. Discontinuously synthesized fragments of DNA are joined by

1) RNA ligase
2) Helicase
3) DNA polymerase II
4) DNA ligase

43. The process of copying genetic information from one strand of DNA into RNA is termed as

1) Translation
2) Transformation
3) Transcription
4) Reverse transcription

44. The promoter site and the terminator site for transcription are located at

1) 3' (downstream) end and 5' (upstream) end, respectively of the transcription unit
2) 5' (upstream) end and 3' (downstream) end, respectively of the transcription unit
3) The 5' (upstream) end
4) The 3' (downstream) end

45. The functional unit of gene that specifies synthesis of one polypeptide is

1) Muton
2) Cistron
3) Intron
4) Transposon

46. Which of the following is not the component of transcription unit?

1) Promoter
2) Terminator
3) Structural gene
4) Inducer

47. Read the following statements and choose the correct option.

Statement 1: The principle of complementarity governs the process of transcription, except the adenosine complements which now base pairs with uracil instead of thymine.

Statement 2: Unlike the process of replication, which once set in, the total DNA of an organism gets duplicated, in transcription only a segment of DNA and only one of the strands is copied into RNA.

1) Only statement 2 is correct but statement 1 is incorrect
2) Only statement 1 is correct but statement 2 is incorrect
3) Both statements are correct
4) Both statements are incorrect

48. Find the odd one out with respect to structure of transcription unit.

1) Promotor
2) Terminator
3) Structural gene
4) m-RNA

49. During splicing, which of the following is removed?
1) Exons
2) Introns
3) hnRNA
4) Primary transcript

Genetic Code and Translation

50. Find the incorrect match with respect to the translation process.
1) rRNA-Catalytic role during peptide bond formation
2) tRNA-Transfer amino acid
3) mRNA-Codons
4) UTR-Only at 5'end

51. Genetic code translates the language of
1) RNA into that of protein
2) RNA into that of DNA
3) Amino acids into that of RNA
4) Amino acids into that of DNA

52. Of the A different possible codons, B specify amino acids and C codons do not specify any amino acids.
1) A-64; B-20; C-3
2) A-64; B-61; C-3
3) A-20; B-15; C-5
4) A-20; B-17; C-3

53. The number of codons that code different amino acids is
1) 64
2) 61
3) 20
4) 63

54. One of the following is true with respect to AUG is
1) It codes for methionine only
2) It is also an initiation codon
3) It codes for methionine in both prokaryotes and eukaryotes
4) All of the above

55. Match the columns and select the correct option

Column I		Column II	
i	UUU	a	Serine
ii	GGG	b	Methionine
iii	UCU	c	Phenylalanine
iv	CCC	d	Glycine
v	AUG	e	Proline

	i	ii	iii	iv	v
1)	c	d	a	e	b
2)	c	a	d	e	b

3) c d e a b
4) b d a e c

56. The process of polymerization of amino acid to a polypeptide is known as
1) Replication
2) Transcription
3) Translation
4) Transduction

57. UTRs in mRNA are
1) Required for efficient translation process
2) Present at 5' end before start codon
3) Present at 3' end after stop codon
4) All of the above

58. How many amino acids will be produced by the given mRNA sequence if the 16th base is substituted by uracil?
5'AUGCGCAAACCCUAGCGA3'
1) Four
2) Five
3) Two
4) Three

59. Degeneration of a genetic code is attributed to the
1) First nucleotide of a codon
2) Second nucleotide of a codon
3) Third nucleotide of a codon
4) Entire codon

60. The following are the salient features of genetic code. Which one is incorrect?
1) The codon is triplet. 61 codons code for amino acids and 3 codons do not code for any amino acids.
2) The codon is read in mRNA in a contiguous fashion, there are no punctuations.
3) The code is always universal without any exception.
4) From bacteria to human UUU would code for Phenylalanine (phe).

61. The amino acid serine (Ser) is coded by four codons. This shows
1) Ambiguous nature of codons
2) Degeneracy of codons
3) Universal nature of codons
4) Dual nature of codons

62. GUG normally codes for valine, but at start position, it codes for methionine. The given statement is an exception to which property of genetic code?
1) Degeneracy
2) Nonoverlapping
3) Ambiguous nature
4) Unambiguous nature

63. In eukaryotes, the, regulation of gene expression occurs at
(A) Transcription level
(B) Processing level
(C) Transport of ribosomal subunits from nucleus to cytoplasm
(D) Translation level
1) Only (A) and (B)
2) Only (D)

3) Only (A), (B), and (D)
4) Only (A), (B) (C), and (D)

64. Operon model of gene regulation in prokaryote was proposed by
1) Jacob and Monod
2) Watson and Crick
3) Beadle and Tatum
4) Francis Crick

65. The predominant site for the control of gene expression in prokaryote is
1) Splicing
2) Transcriptional initiation
3) Translational level
4) Processing level

66. Match the column I with column II for genes of lac operon, and select the correct option.

Column I		**Column II**	
A	z gene	i	Transacetylase
B	y gene	ii	Repressor
C	a gene	iii	β -galactosidase
D	i gene	iv	Permease

1) A-iv; B–iii; C–i; D-ii
2) A-iii; B-i; C–iv; D-ii
3) A-iii; B-iv; C-i; D-ii
4) A-iii; B-ii; C-iv; D-i

67. In lac operon, the regulator gene is denoted as i-gene. Here the term 'i' denotes for
1) Initiator
2) Inhibitor
3) Inducer
4) Introducer

68. Which of the following is the function of enzyme coded by Lac y gene?
1) Hydrolysis of lactose
2) Transportation of lactose into the cell
3) Induction of repressor protein
4) Breakdown of glucose

69. Read the following statements and choose the correct option.

I. The genes in a cell are expressed to perform a particular function or a set of functions.
II. The bacteria use glucose and galactose as a source of energy.
III. In most other operons as well, the genes present in the operon are needed together to function in the same or related metabolic pathway.
IV. Regulation of lac operon can be visualised as regulation of enzyme synthesis by its substrate.

Correct statement(s) is/are
1) Only I, III and IV
2) All of the above
3) Only I, II and III
4) Only II, III and IV

70. Read the following statements and choose the correct option.

I. The repressor protein binds to the operator region of the operon and prevents RNA polymerase from transcribing the operon.
II. Glucose or galactose can also act as inducers for lac operon.

III. A very high level of expression of lac operon has to be present in the cell all the time, otherwise lactose cannot enter the cells.

IV. In lac operon, the lactose induces the operon.

Correct statement(s) is/are

1) Only I and IV
2) Only II and III
3) Only II and IV
4) All of the above

71. Read the following statements with respect to lac operon, and select the correct option.

A. It has three structural genes.
B. Glucose can act as inducer.
C. Repressor protein is acting in presence of lactose.
D. Repressor is synthesized by regulator gene.

1) Only (B) is incorrect.
2) (A) and (C) are correct.
3) (A) and (D) are correct.
4) (A), (B), and (D) are correct.

72. ß-galactosidase is synthesized by E. coli to catalyze hydrolysis of A into B and C

1) A-Lactose; B-Glucose; C-Galactose
2) A-Lactose; B-Glucose; C-Fructose
3) A-Galactose; B-Glucose; C-Fructose
4) A-Lactose; B-Maltose; C-Glucose

Human Genome Project and DNA Fingerprinting

73. The Human Genome Project was coordinated by

1) U.S. Department of Energy
2) National Institute of Health
3) Wellcome Trust of UK
4) All of the above

74. Human Genome Project (HGP) was closely associated with the rapid development of a new area in biology known as

1) Biotechnology
2) Bioinformatics
3) Biolistics
4) Genetic Engineering

75. Select the most commonly used vector in the Human Genome Project.

1) Bacteriophage
2) Bacteriophages
3) Plasmids only
4) BAC and YAC

76. Match the columns and select the correct option.

Column I		Column II	
a	DNA fingerprinting	i	Variation at genetic level
b	Largest human gene	ii	Alec Jeffreys
c	Bioinformatics	iii	Human Genome Project
d	Polymorphism	iv	Dystrophin

a b c d

1) 2 4 1 3
2) 3 1 2 4
3) 2 3 4 1
4) 2 4 3 1

77. Given below are some of the important goals of HGP.

I. Identify approximately all the 30,000-40,000 genes in human DNA.
II. Determine the sequences of the 6 billion chemical base pairs that make up human DNA.
III. Store this information in databases.
IV. Improve tools for data analysis.
V. Transfer related technologies to other sectors, such as industries.
VI. Address the ELSI that may arise from the project.

Which one is incorrect?

1) Only I and II
2) Only I, IV and V
3) Only II and IV
4) None of the above

78. Read the following statements and choose the correct option.

Statement 1: Knowledge about the effects of DNA variations among individuals can lead to revolutionary new ways to diagnose, treat, and someday prevent the thousands of disorders that affect human beings.

Statement 2: Besides providing clues to understanding human biology, learning about non-human organisms DNA sequences can lead to an understanding of their natural capabilities that can be applied toward solving challenges in health care, agriculture, energy production, environmental remediation.

1) Only statement 2 is correct but statement 1 is incorrect
2) Only statement 1 is correct but statement 2 is incorrect
3) Both statements are correct
4) Both statements are incorrect

79. The strategies of Human Genome Project include

1) Mapping
2) Sequencing
3) Functional analysis
4) All of the above

80. Sequencing the whole set of genomes that contains all the coding and non-coding sequence and assigning different regions in the sequence with functions is termed as

1) Sequence annotation
2) Expressed sequence tags
3) Single-nucleotide polymorphism
4) Expressed sequence database

81. Read the given statements of assertion and reason and select the correct option.

Assertion: Repetitive sequences are stretches of DNA sequences that are repeated many times, sometimes hundred to thousand times.

Reason: They are thought to have no direct coding functions, but they shed light on chromosome structure, dynamics, and evolution.

1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
3) If assertion is true, but reason is false.
4) If both assertion and reason are false.

Answer Key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
4	1	2	2	3	4	1	3	3	2	3	4	4	3	4
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
4	3	2	1	4	2	4	3	3	3	2	4	1	2	4
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
2	1	2	3	4	4	3	4	3	3	4	4	3	2	2
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
4	3	4	2	4	1	2	2	4	1	3	4	1	3	3
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**	**71**	**72**	**73**	**74**	**75**
2	4	3	1	2	3	2	2	2	1	3	1	4	2	4
76	**77**	**78**	**79**	**80**	**81**									
4	1	3	4	1	2									